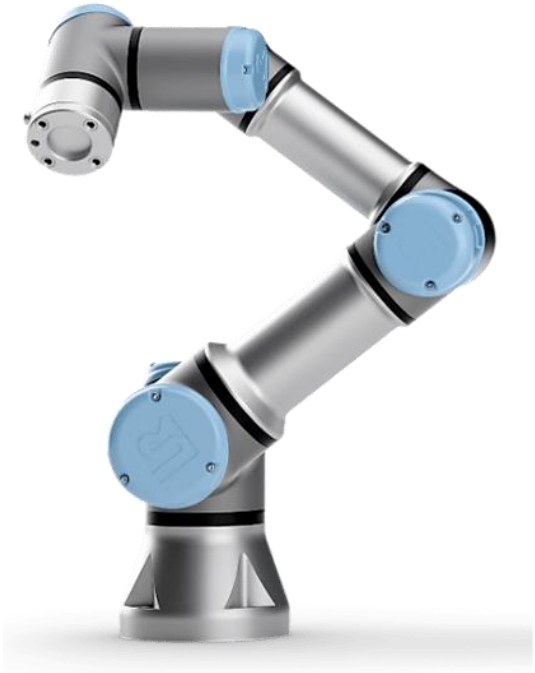
Experimental Evaluation of Low-Cost Robotic Arm Repeatability Using Optical Tracking

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M3 Report Presentation

Last Revisited: May 4, 2026

Advances in physical intelligence have expanded the role of robotics in research and education, however, the high costs can make it unfavorable.



PR-UR-003-0000

Universal Robots UR3e Collaborative Robot Arm

Universal Robots

$33,011 USD

Ships in 17 days

Img Src: https://vention.io/parts/universal-robots-ur3e-collaborative-robot-arm-2446



LeRobot SO-100 Robot Arms

Price: ≈ $230 USD

Img Src: https://github.com/TheRobotStudio/SO-ARM100

Design and implement an experimental system to quantify robotic arm repeatability, precision, and trajectory deviation.

Define

Performance Metrics

- **Repeatability**
  Consistency across trials
- **Precision**
  Deviation of results
- **Trajectory Deviation**
  Follower vs Leader Arm

Measure

- **Multi Trial Testing**
- **Deviation across runs**
- **Leader vs Follower Tracking**

Outcomes

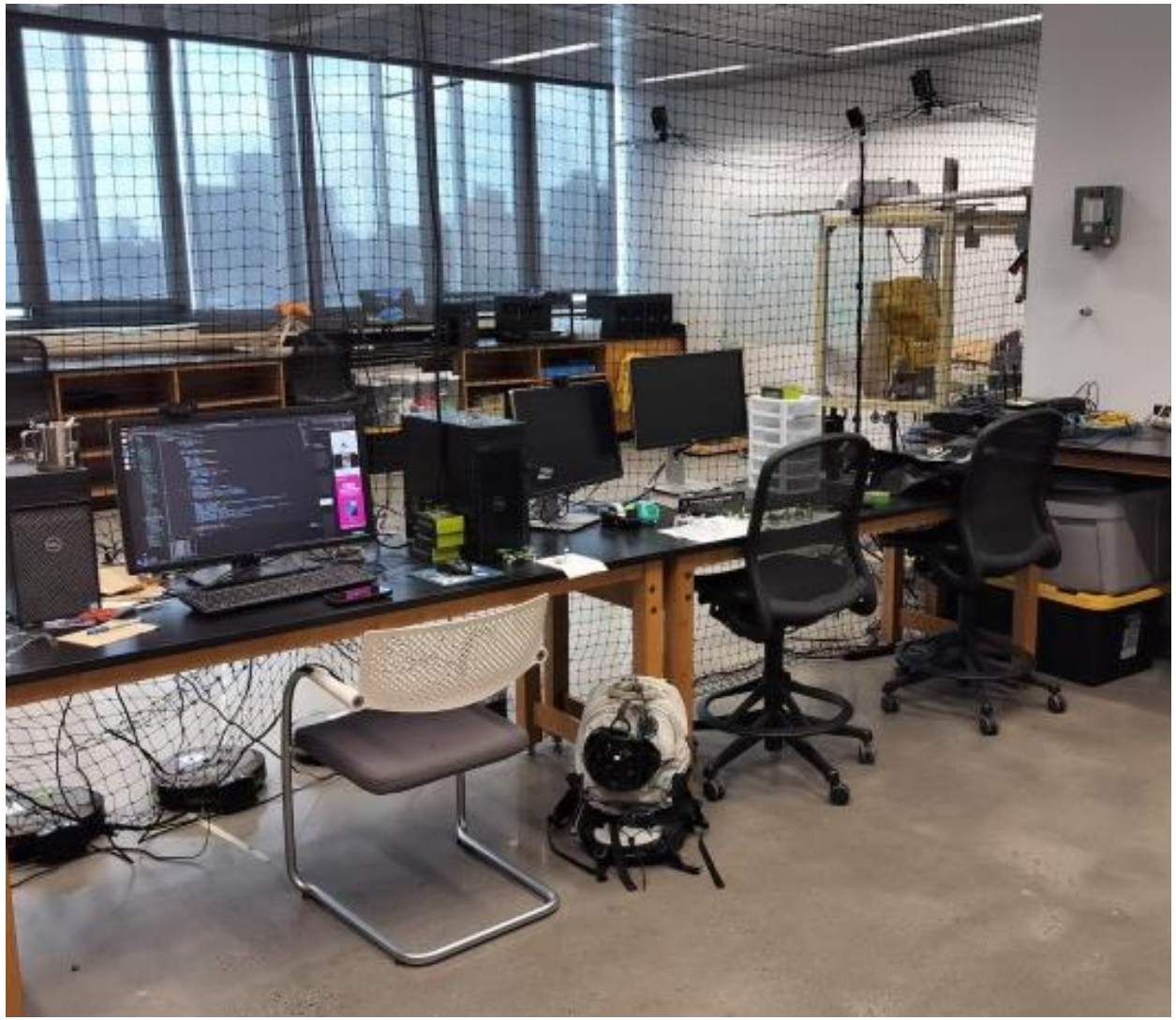
Performance Metrics

Variability Across Units

Scalability

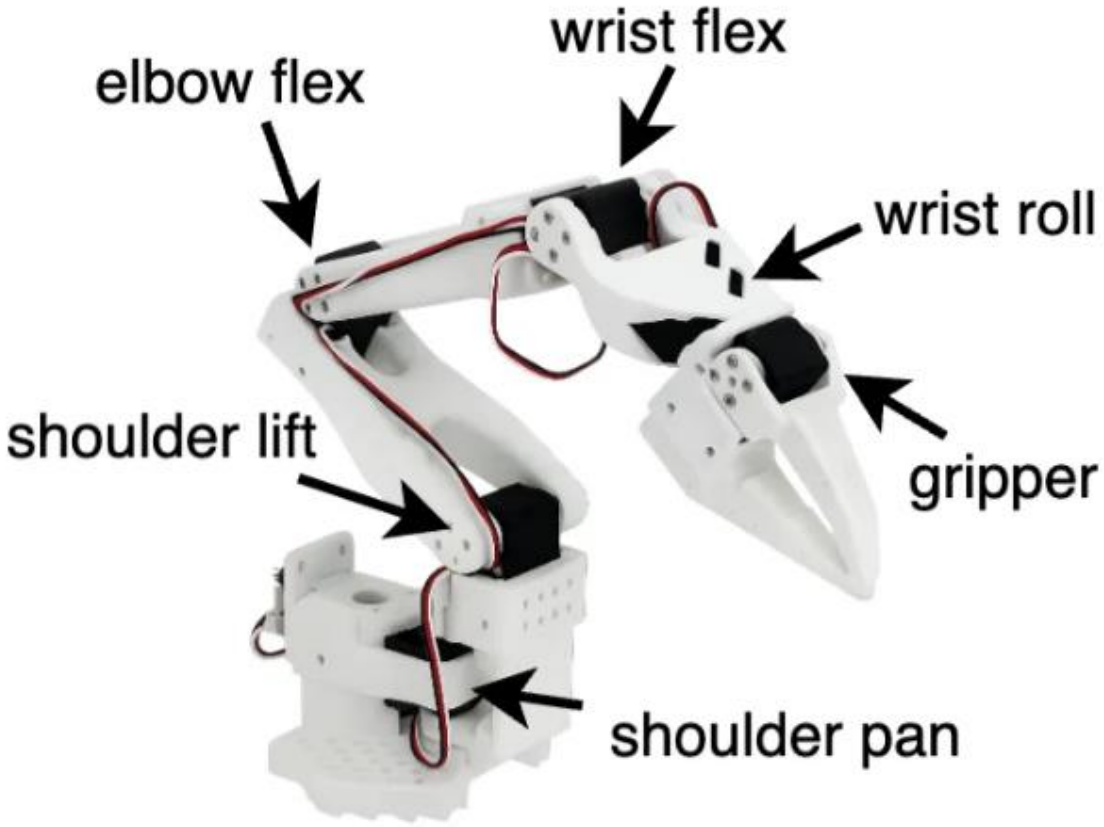
An optical tracking system allows for holistic analysis of compliance, like the one in the Dynamics and Controls Lab (41 CS RM 705)

- The Vicon Vantage system has the following specifications:
  - Accuracy range of ± 0.20 mm
  - 240 samples per second
  - Uses IR and reflective balls to track position
  - Uses Vicon Tracker 3.10
- 3D positional data allows for computation of relative joint position (angles)



Img Src: https://cooper.edu/sites/default/files/styles/820w_16_9/public/2024-07/705image1.jpg?itok=ICguGIzX

By attaching 4 reflective markers to each link, the robot can be tracked using the Vicon system.



Robot Arm Links and Labeled Motors

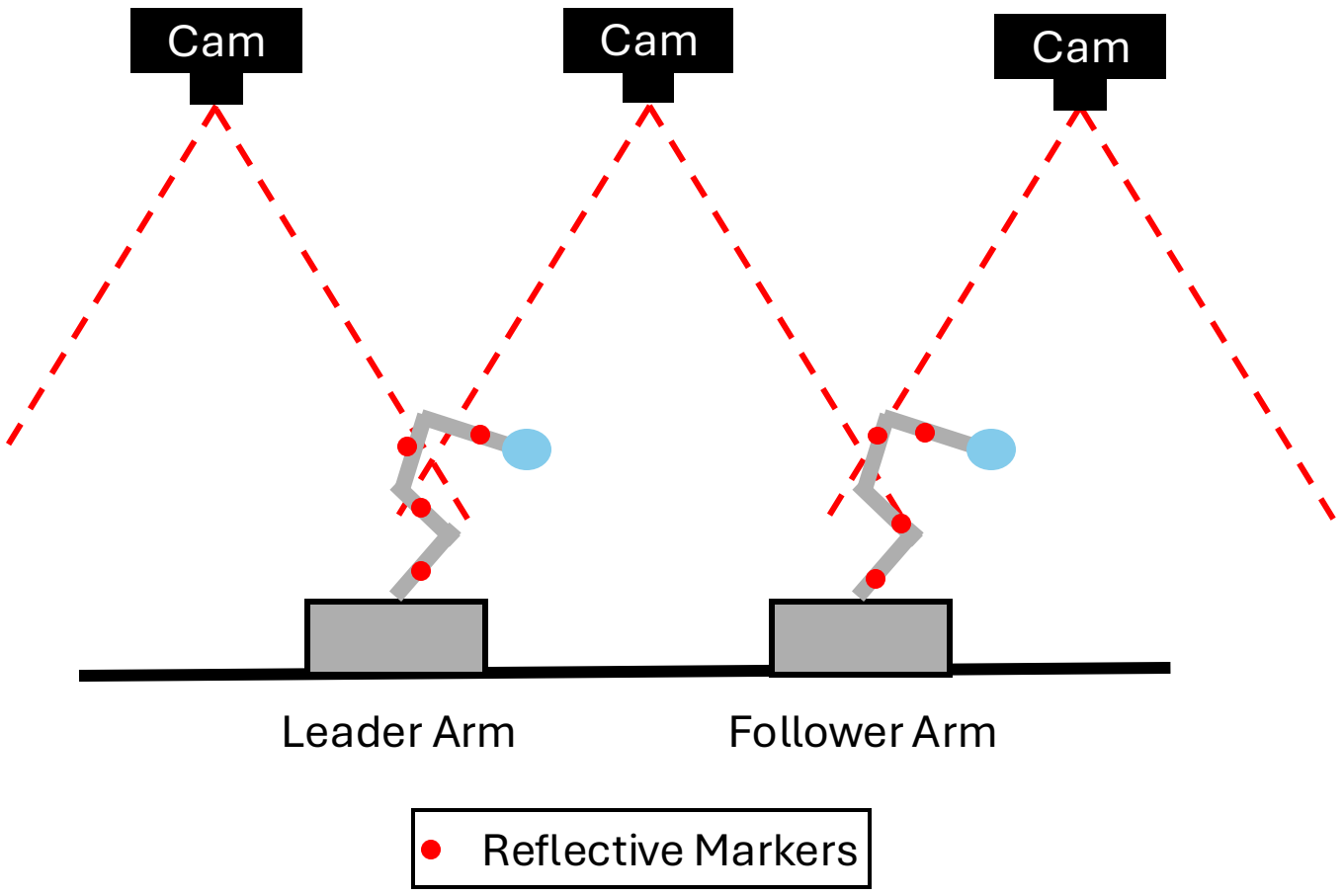
Img Src: https://huggingface.co/spaces/lerobot/robot-learning-tutorial



Precision Pearl Reflective Markers

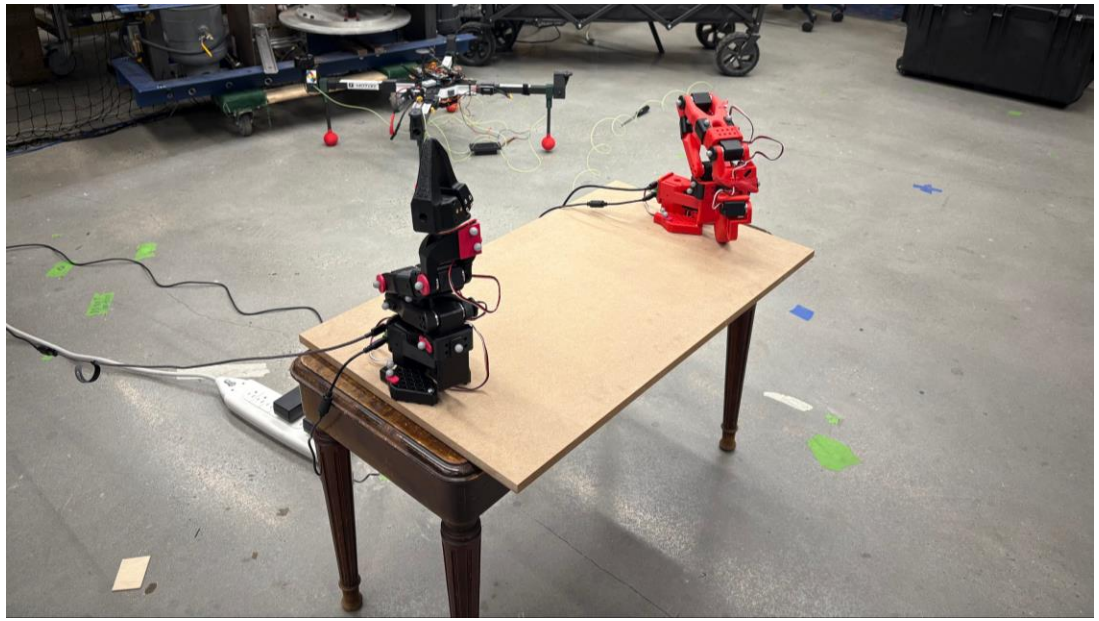
Img Src: https://bleng.com/collections/high-precision-pearl-reflective-markers

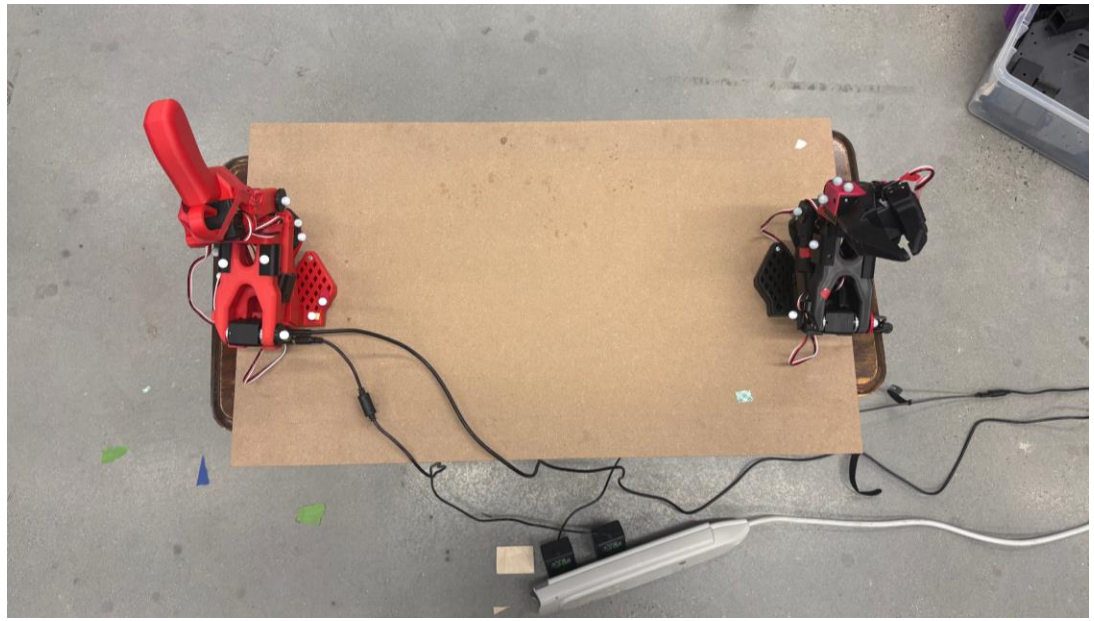
By setting up one leader arm and one follower arm within the field of view of the Vicon system and following standards, data on deviations can be obtained.



- Geometry created by the 4 reflective balls must not be identical on any link.
- Reflective balls cannot be placed too close together or system cannot tell them apart.
- The four reflective balls on each link cannot sit on the same plane.

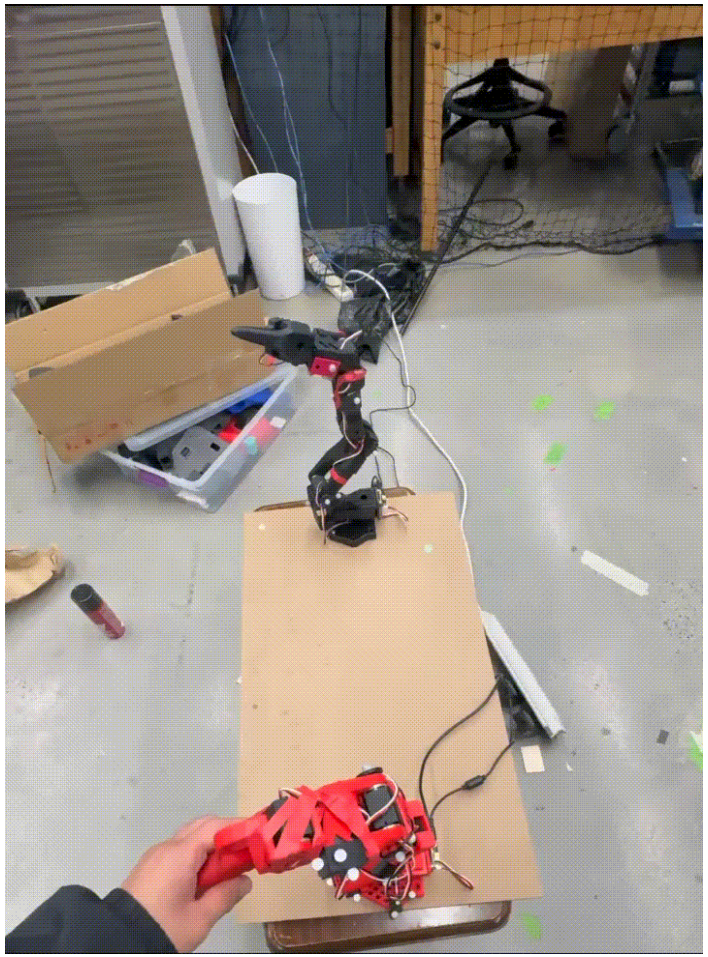
The arms were set up in a manner at which the base of both arms are a fixed distance apart without interfering with each others' movements.

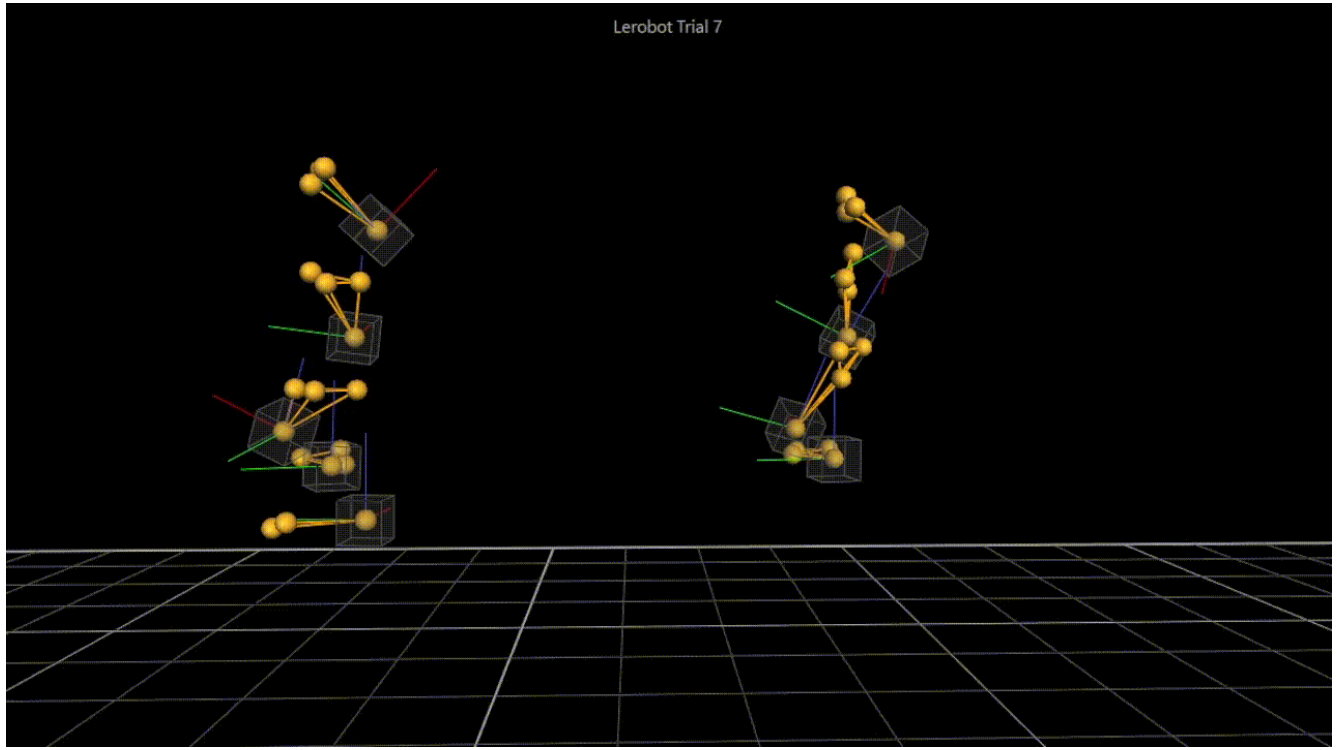




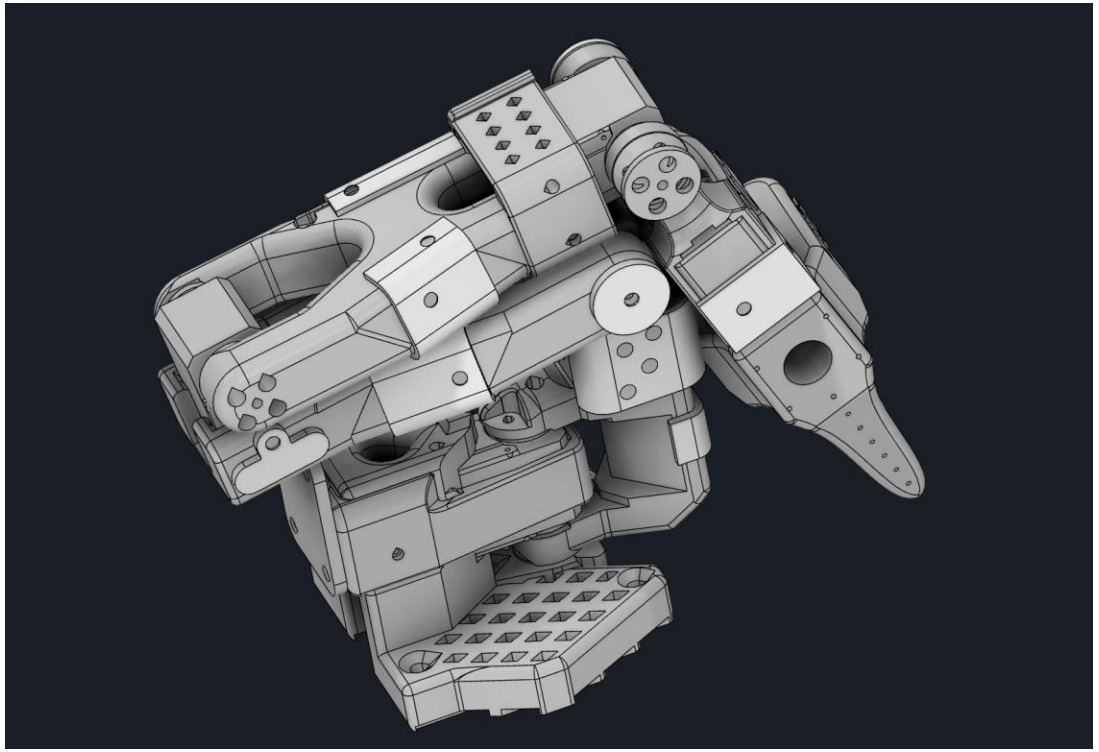
Placed one ball per link in the same spot on each arm to use as a local frame, referred to as the origin ball.

Arms were operated for multiple trials for about 3 minutes per trial.





Tracking balls were placed through repeatably indexed clips



- A series of clips were designed to allow for up to 8 tracking balls to be placed per linkage
- All ball placement points were designed to be unique and sufficiently spaced for differentiation
- 1 tracker ball was placed precisely at each joint origin to allow for flexible setups within the tracker software

By using the Vicon Tracker, the position and rotation with respect to the Vicon global frame of each link is obtained.

Want to have deviation in position between follower and leader arms, so will only get deviations when the arms are stopped.

$$t = \frac{Frame - First\ Frame}{sample\ rate}$$

A stop is considered when the robot does not move for 3 seconds.

What is important is the position of the link relative to the base of the arm, not the global frame, the changes in position are obtained by

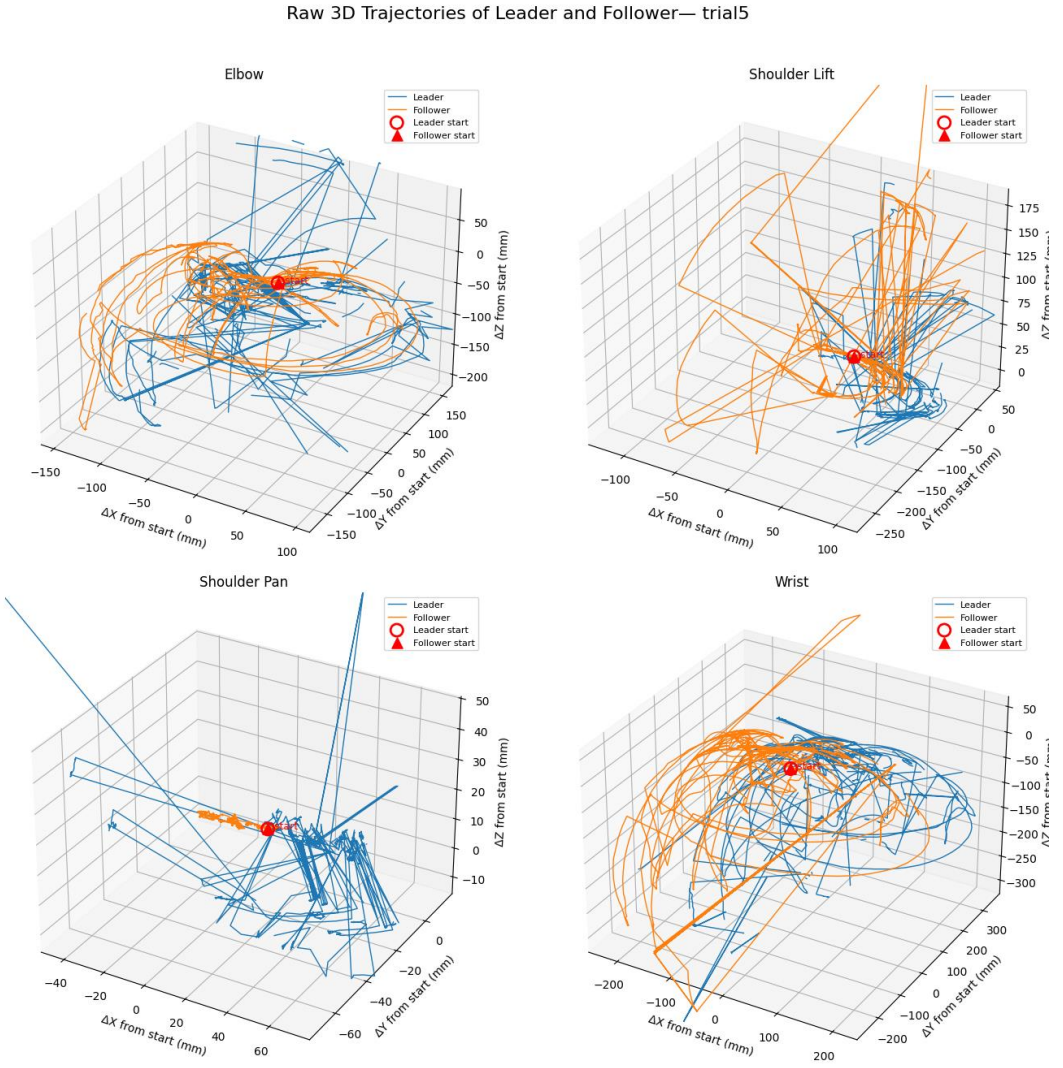
$$\Delta p(t) = p(t) - p_0$$

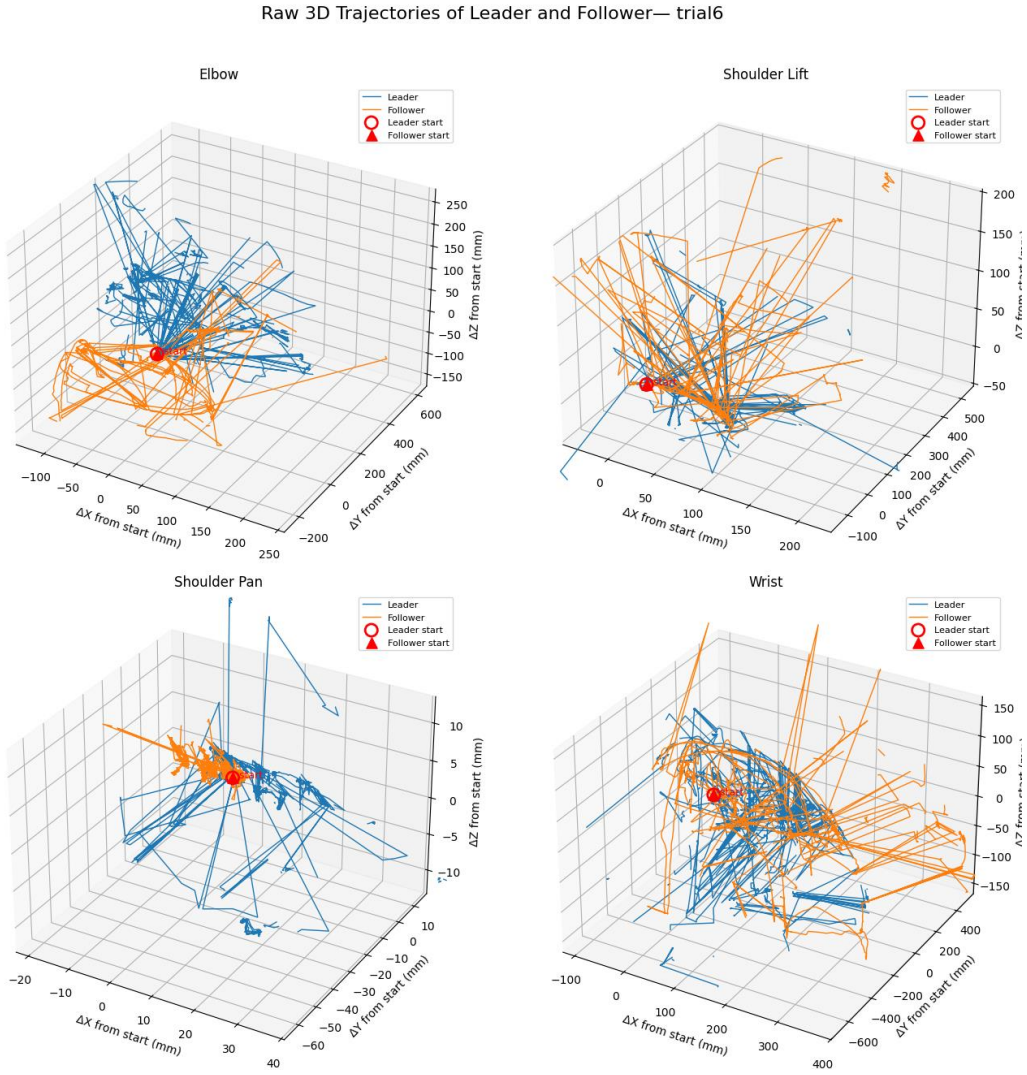
Then, the main comparison ends up being

$$\Delta p_{Follower} - \Delta p_{Leader}$$

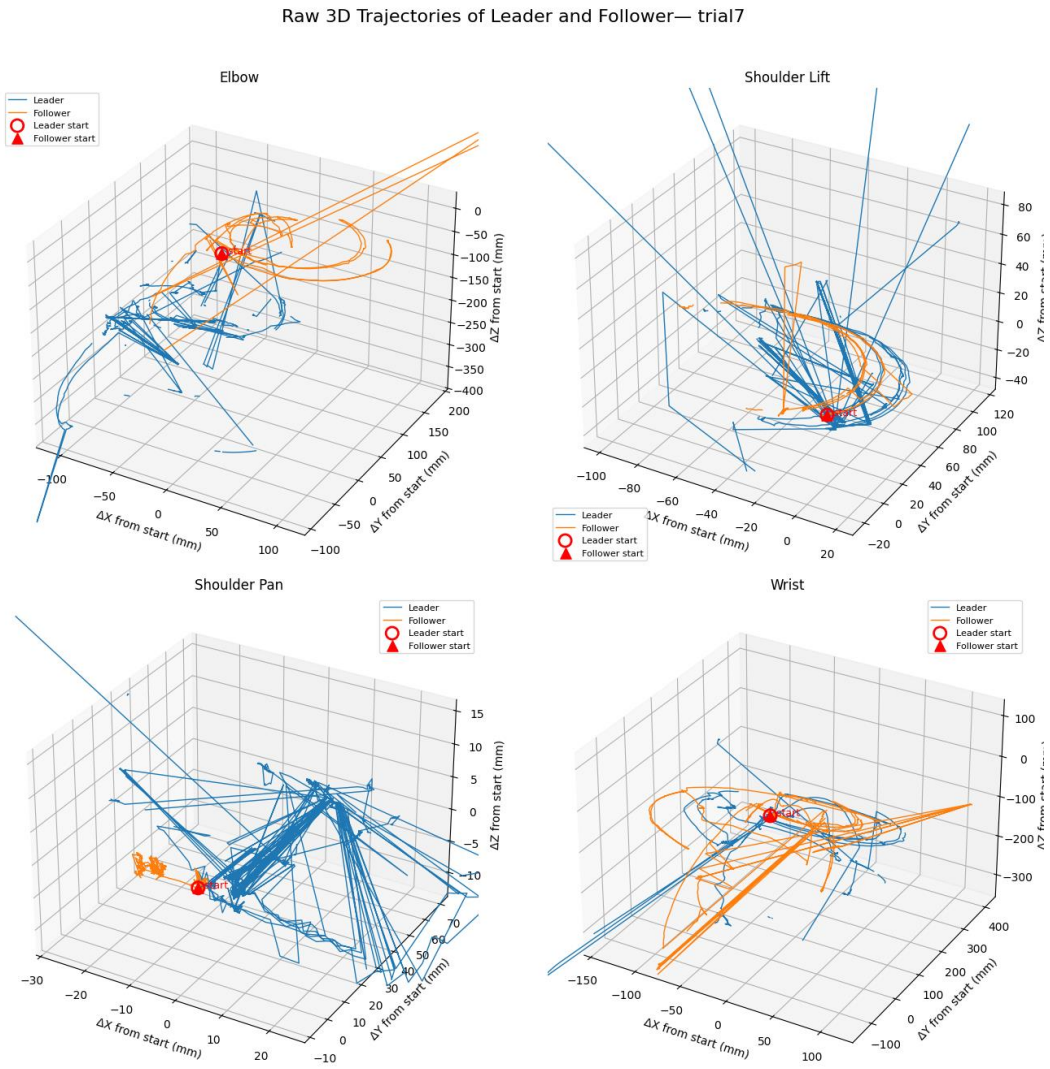
Then, clean the data.

Raw motion data indicates significant tracking noise and inconsistent follower response

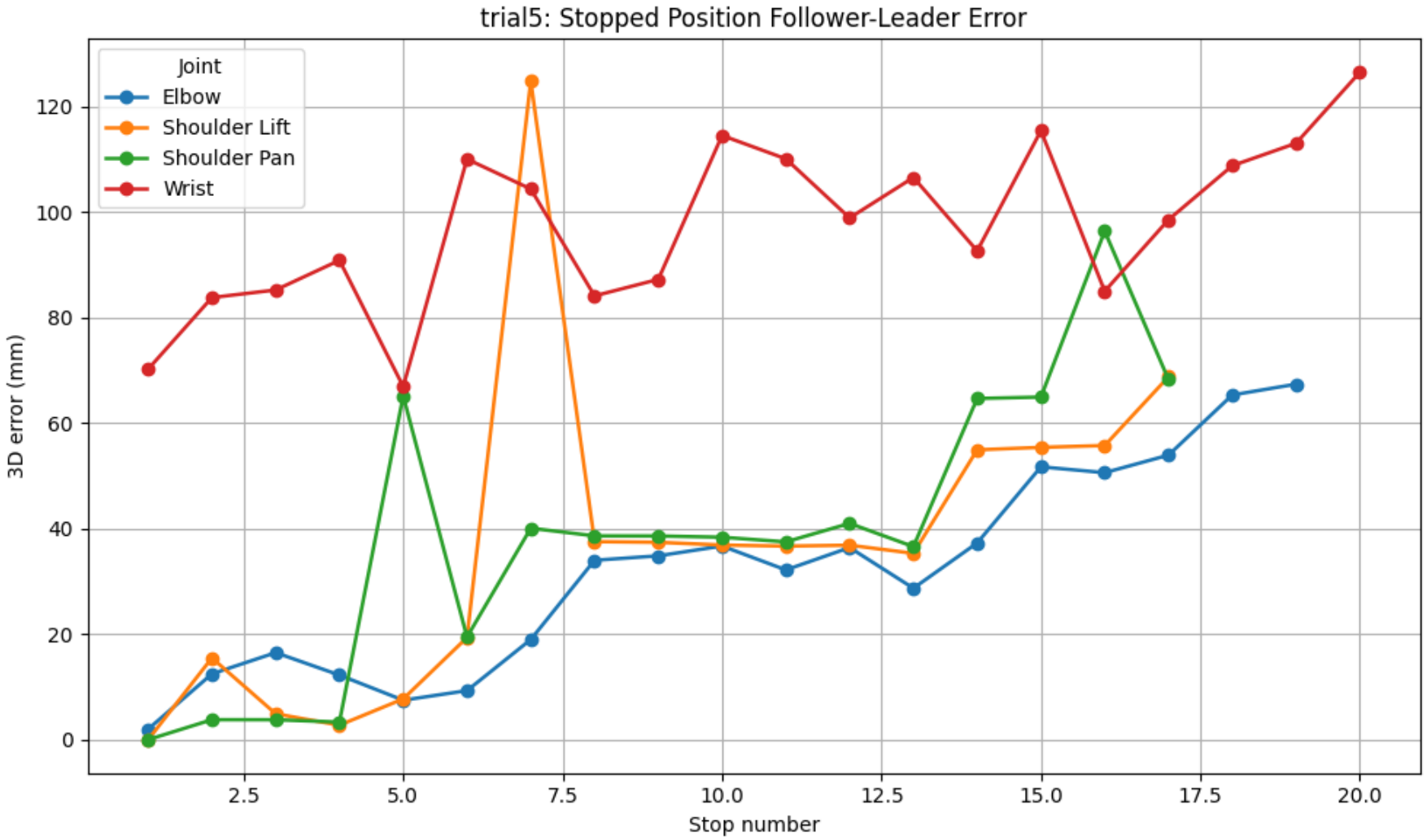




Raw motion data indicates significant tracking noise and inconsistent follower response



Follower arm show increasing error across stops, with wrist showing the largest deviation



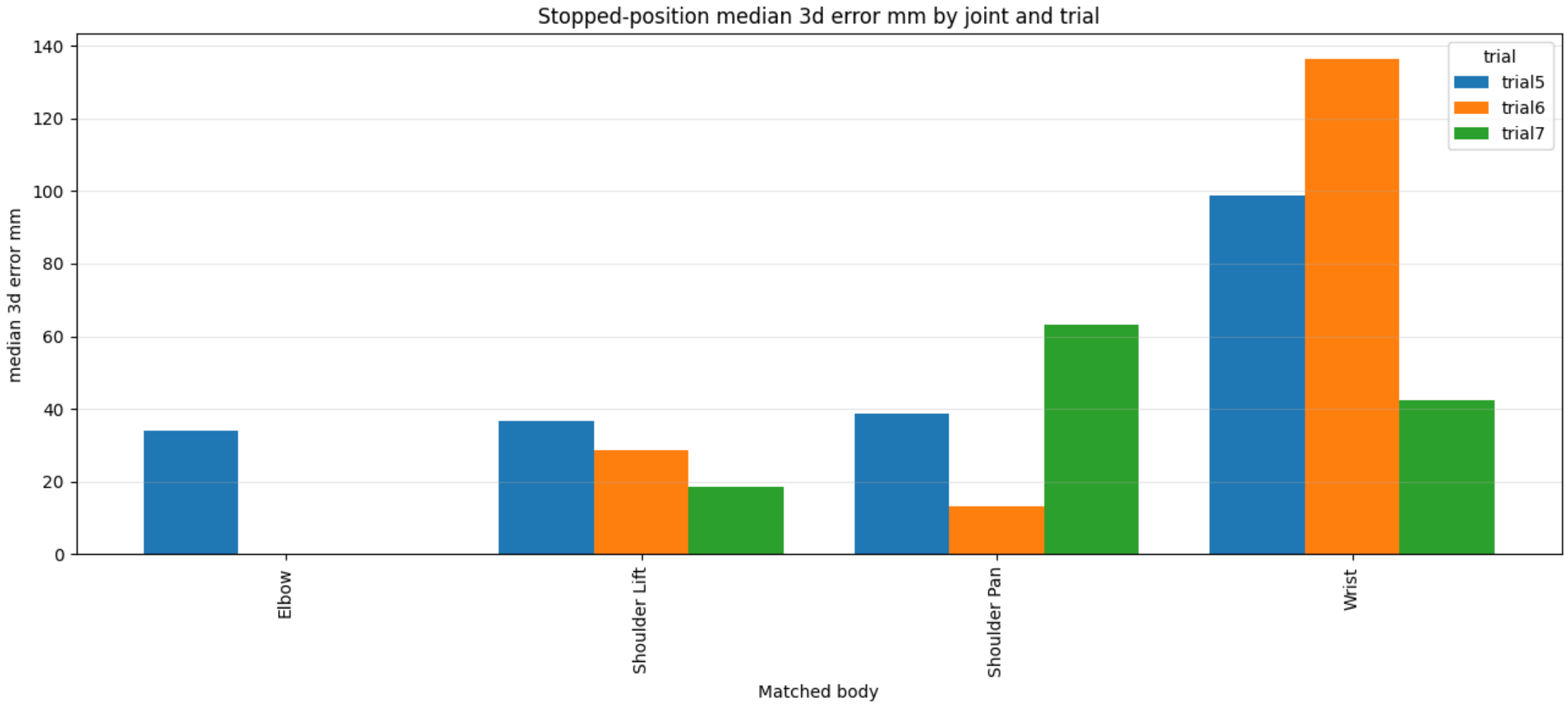
Wrist error remains consistently high while shoulder lift shows large variability across stops



Improved stability observed, but wrist remains the primary source of error



Follower error increases toward the wrist



Demonstrated a Vicon-based method to measure SO-100 motion while identifying the need for stronger tracking

- Compared leader and follower motion using start-relative position changes to account for separate arm locations in the capture volume.
- Used stopped-position comparisons to evaluate how closely the follower reached the leader's settled positions.
- Found that deviation generally increased toward the wrist/end-effector, indicating accumulated error along the arm.
- Established a workable measurement approach, while showing that improved marker tracking are needed for more reliable accuracy results.

Improving tracking reliability will strengthen future SO-100 motion measurements

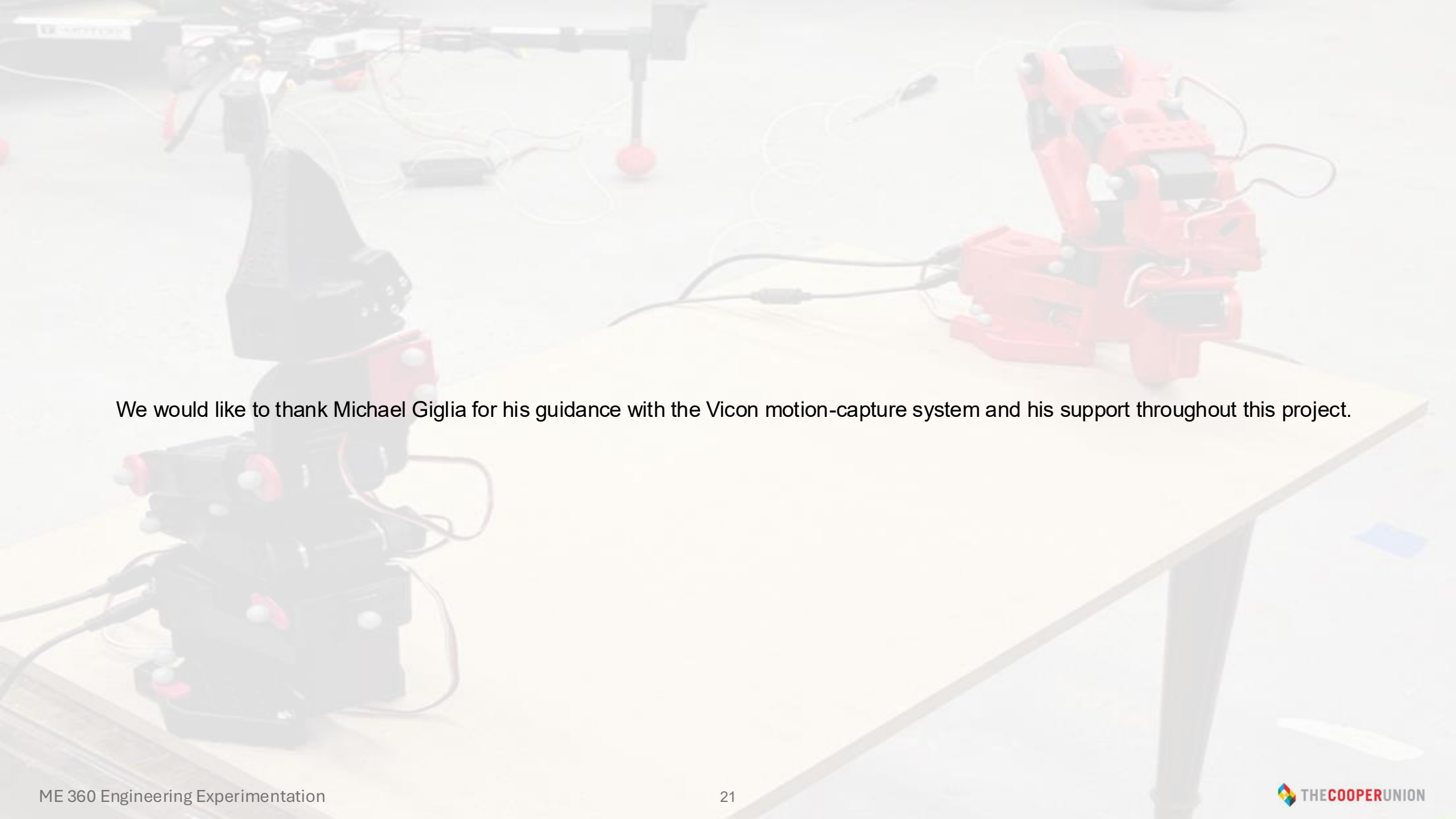
- Improve camera coverage to increase tracking reliability
- Increase camera visibility around the wrist and elbow to reduce marker occlusion in high-error regions.
- Use autonomous scripted arm motions to create repeatable trajectories across trials.
- Redesign robot parts with integrated tracking-ball slots for more consistent rigid-body tracking.
- Optimize marker-cluster geometry by spacing balls farther apart in distinct 3D shapes to prevent marker swapping and shifting.

A more robust system is required for higher quality data

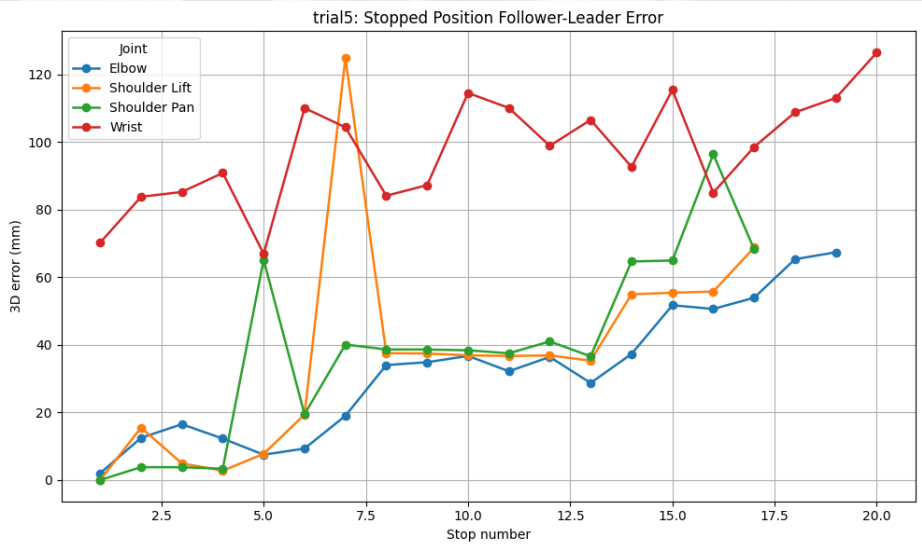
- Motion capture through optical tracking shows significant promise
- Data when captured is of generally high quality
- The robustness of the system must be improved for better results

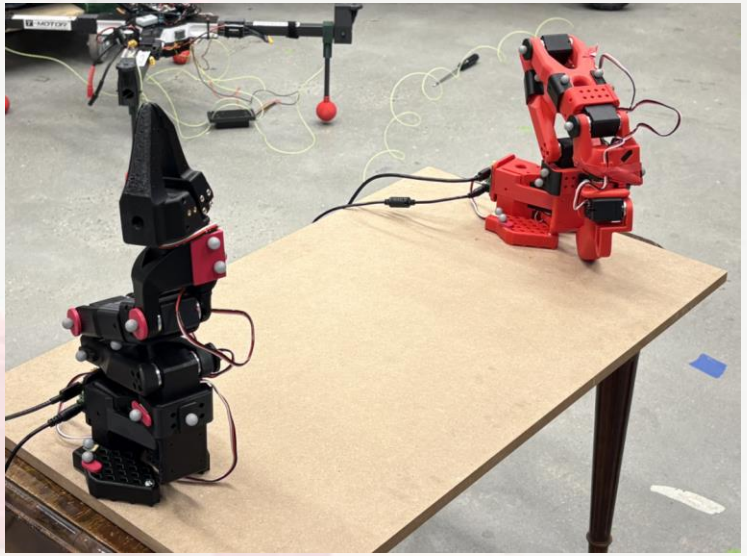
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- https://media.licdn.com/dms/image/v2/D5612AQGW-op3Ew2Uuw/article-cover_image-shrink_720_1280/B56ZcDHgZVHUAQ-/0/1748103989523?e=2147483647&v=beta&t=uArUMYDanIQjEH7ksSPtLmcdHbN1KEz7mT6M2JK-0nM

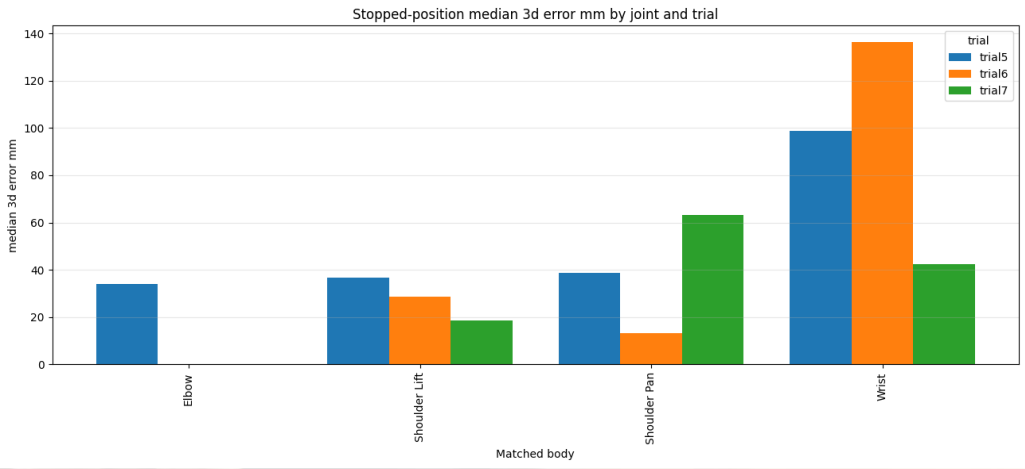


We would like to thank Michael Giglia for his guidance with the Vicon motion-capture system and his support throughout this project.


trial5: Stopped Position Follower-Leader Error
Joint
Elbow
Shoulder Lift
Shoulder Pan
Wrist
3D error (mm)
Stop number

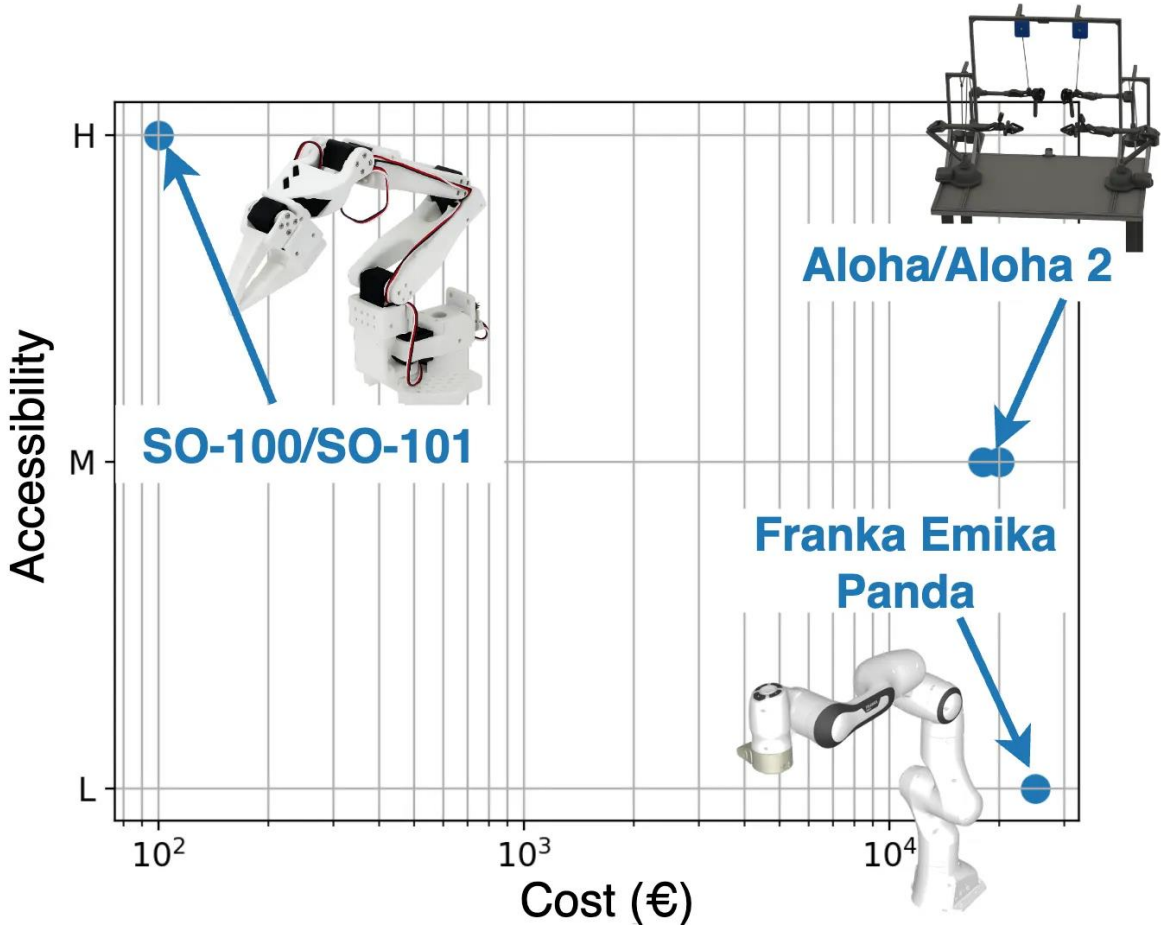


Thank you

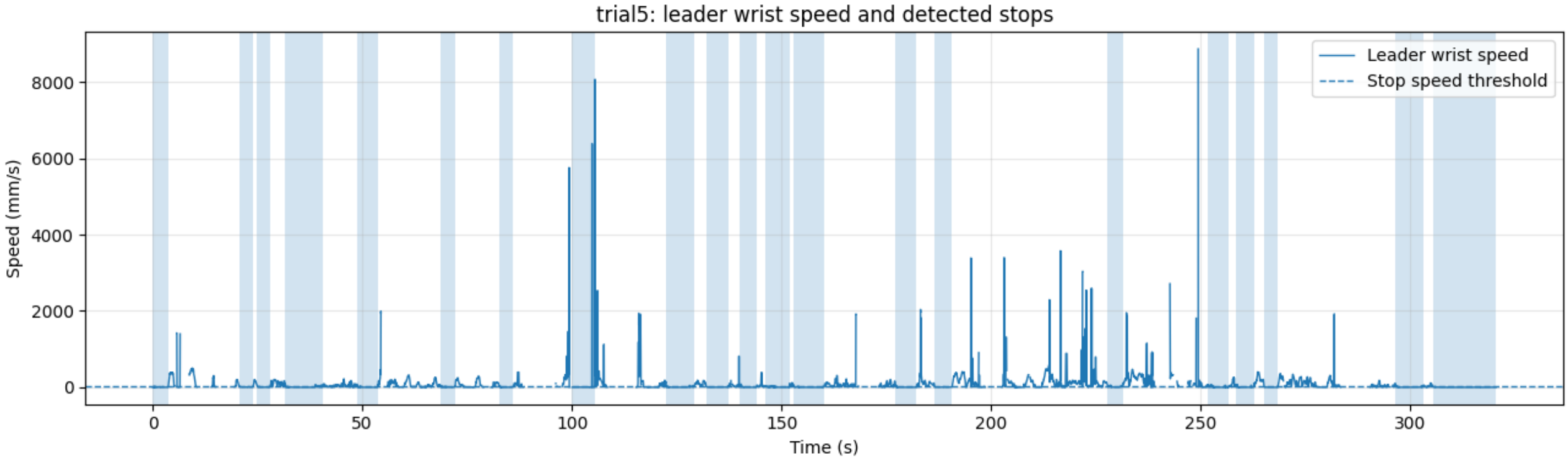

Stopped-position median 3d error mm by joint and trial
trial
trial5
trial6
trial7
median 3d error mm
Elbow
Shoulder Lift
Shoulder Pan
Wrist
Matched body



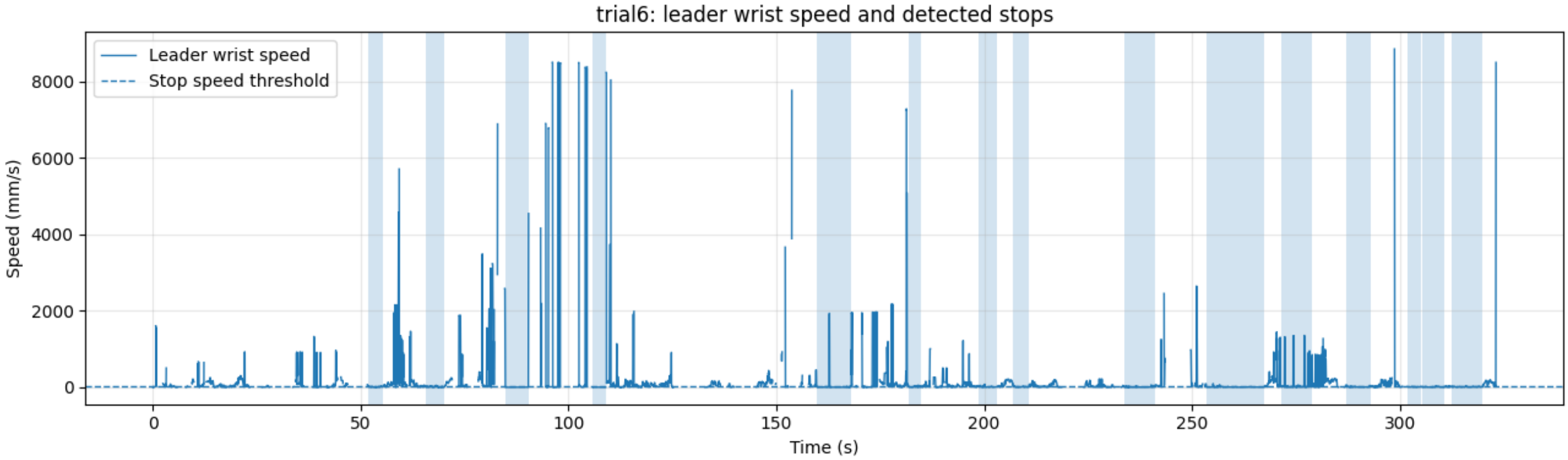
[Appendix] SO-100 Market Positioning



[Appendix] Speed – Trial 5



[Appendix] Speed – Trial 6



[Appendix] Speed – Trial 7

